CAPSTONE PROJECT DESIGN DOCUMENT

PS-091: Digital Ad Auction, Budget Pacing and Frequency-Cap Engine

Team ID: T049 | **Domain**: Media - AdTech

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Placement Coaching Capstone Project — 10 SECONDS

1. Domain Glossary

Term	Definition & Domain Significance
Publisher	The digital property supplying ad slots. Sets a floor CPM reserve price.
Advertiser	Brand purchasing ad space. Holds either a PREPAID wallet or CREDIT account.
Campaign	An ad initiative with targeting, flight dates, bid model, and daily budget.
CPM	Cost Per Mille (per 1,000 views). Billed immediately on impression delivery.
CPC	Cost Per Click. Billed only when an impression receives a verified click.
eCPM	Effective CPM: Standardised revenue per 1,000 impressions: `eCPM = CPC * CTR * 1000`.
Floor Price	Reserve CPM set by publisher. Requests clear at or above this price.
Second-Price Auction	Highest eligible bidder wins; clears at `max(2nd_eCPM, Floor) + Rs 0.01`. Single bidder pays Floor.
Frequency Cap	Max 3 impressions per user per calendar day and max 1 per clock hour.
Budget Pacing	Spreads daily budget across 24 hourly buckets matching the traffic profile.
Parity Throttling	When OVER_PACE (≥120%), campaign sits out on odd request sequence numbers.

Invalid Traffic (IVT)	>5 clicks from same user on same campaign within 10 minutes marks all burst clicks INVALID.
Clawback	Publisher share credited on invalid clicks is clawed back from monthly payout.
Payout Rollover	Monthly payouts under Rs 5,000 roll forward to the subsequent month.

2. Business Rules Register (BR-01 to BR-12)

Rule ID	Rule Name	Operational Logic & Formulas
BR-01	Eligibility & Targeting	Matches geo, device, category. Request time within flight dates. Daily spend < daily budget. Prepaid wallet ≥ 1 max bid; Credit non-suspended.
BR-02	Frequency Capping	Max 3 impressions/user/day (calendar day IST) and max 1 impression/user/hour (clock hour IST).
BR-03	Bid Normalisation	CPM: eCPM = max_bid. CPC: eCPM = max_bid * trailing_CTR * 1000. Trailing CTR default = 0.5% if <1000 impressions; else valid_clicks / impressions.
BR-04	Second-Price Auction	Winner clears at `max(second_eCPM, floor) + 0.01`. Single bidder pays floor. Unfilled if all bids < floor.
BR-05	Auction Tie-Break	Equal eCPM bids won by campaign with lower cumulative spend today, then campaign_id ascending.
BR-06	Budget Pacing	Hourly budget buckets. OVER_PACE if spend ≥ 120% of schedule. In UNDER_PACE, catch-up allows up to 150% of current hour bucket.
BR-07	Pacing Throttling	When OVER_PACE, campaign bids only on even request sequence numbers (`seq_no % 2 == 0`).
BR-08	Billing Mechanics	CPM billed per impression at clearing CPM / 1000. CPC billed on click at max_bid capped at daily budget + 1 bid.
BR-09	Wallet & Credit Limits	Prepaid debits wallet to 0. Credit balance may go down to -credit_limit. Any breaching charge stands and suspends account.
BR-10	IVT Detection & Clawback	>5 clicks in 10-minute rolling window marks all INVALID. Advertiser refunded; publisher share clawed back.

BR-11	Publisher Revenue Share	70% share if monthly billed > Rs 1,00,000; else 60%. Payouts < Rs 5,000 roll to next month.
BR-12	Flight End & Expiry	Campaigns serve until 23:59:59 on flight_end. Unspent budget lapses and never carries forward.

3. Planted Edge Cases Verification (EC-01 to EC-08)

Code	Planted Scenario	Engine Verification	Status
EC-01	Single bidder equals floor	REQ-000466: CMP-001 bids 150.00 = floor 150.00 → pays 150.00 (Rs 0.15/imp)	**VERIFIED**
EC-02	2nd bid below floor	REQ-000171: CMP-004 (180.00) vs CMP-005 (90.00) on PUB-4 (120.00) → clears at 120.01	**VERIFIED**
EC-03	1000 trailing impressions	CMP-006 switches from 0.5% to 0.7% CTR → eCPM 175.00 (REQ-000087)	**VERIFIED**
EC-04	120.0% pacing boundary	CMP-001 throttled on odd REQ-000519 (ThrottledUnfilled=1), bids at hour 1	**VERIFIED**
EC-05	Day/hour frequency boundary	REQ-000463 served (3rd today), REQ-000464 capped, REQ-000465 served at 00:00:00	**VERIFIED**
EC-06	6th click within 10:00	CLK-00057..62 marked INVALID; Rs 90.00 clawed back from PUB-4	**VERIFIED**
EC-07	Publisher at exactly Rs 1,00,000	PUB-5 reaches Rs 1,00,000.00 → stays at 60% revenue share tier	**VERIFIED**
EC-08	Credit breach by Rs 0.01	ADV-07 balance hits -50,000.01 → charge stands, suspended; 12 spiritual stay unfilled	**VERIFIED**